

Curvature and Torsion in the Continuum Theory of Lattice Defects

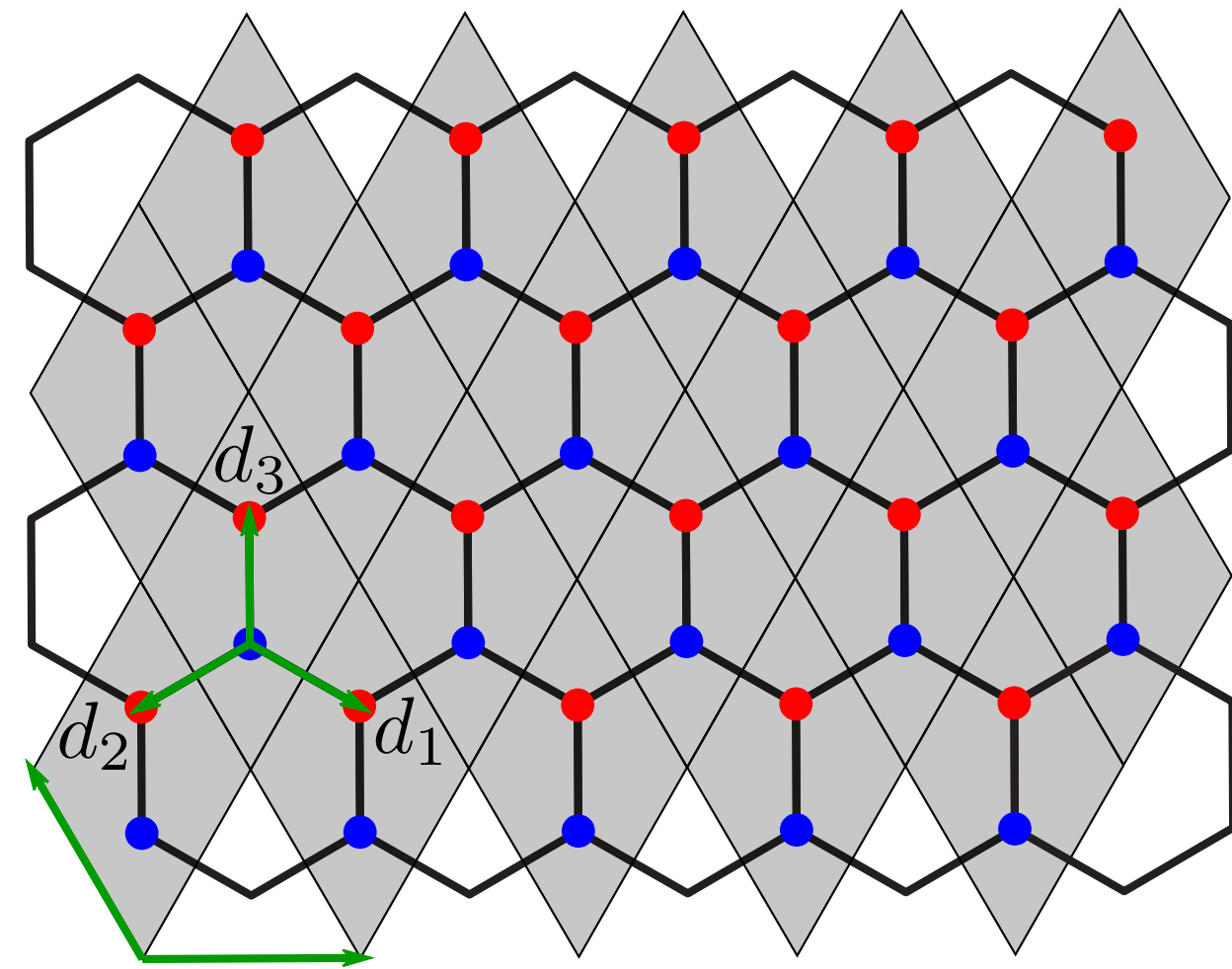
Marvin Henke, Dr. Nikodem Szpak

Faculty of Physics, University Duisburg-Essen, 47048 Duisburg

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Continuum Theory - Graphene



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^3 (a_{\vec{r}}^{\dagger} b_{\vec{r}+\vec{d}_j} + b_{\vec{r}+\vec{d}_j}^{\dagger} a_{\vec{r}})$$

Fourier transform

$$i\partial_i \vec{\psi}(\vec{k}) = \mathbf{h}(\vec{k}) \vec{\psi}(\vec{k})$$

$$\mathbf{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^3 e^{i\vec{k} \cdot \vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \implies \mathbf{h}(\vec{k}) \approx -\frac{3}{2} \delta\vec{k} \cdot \vec{\sigma}$$

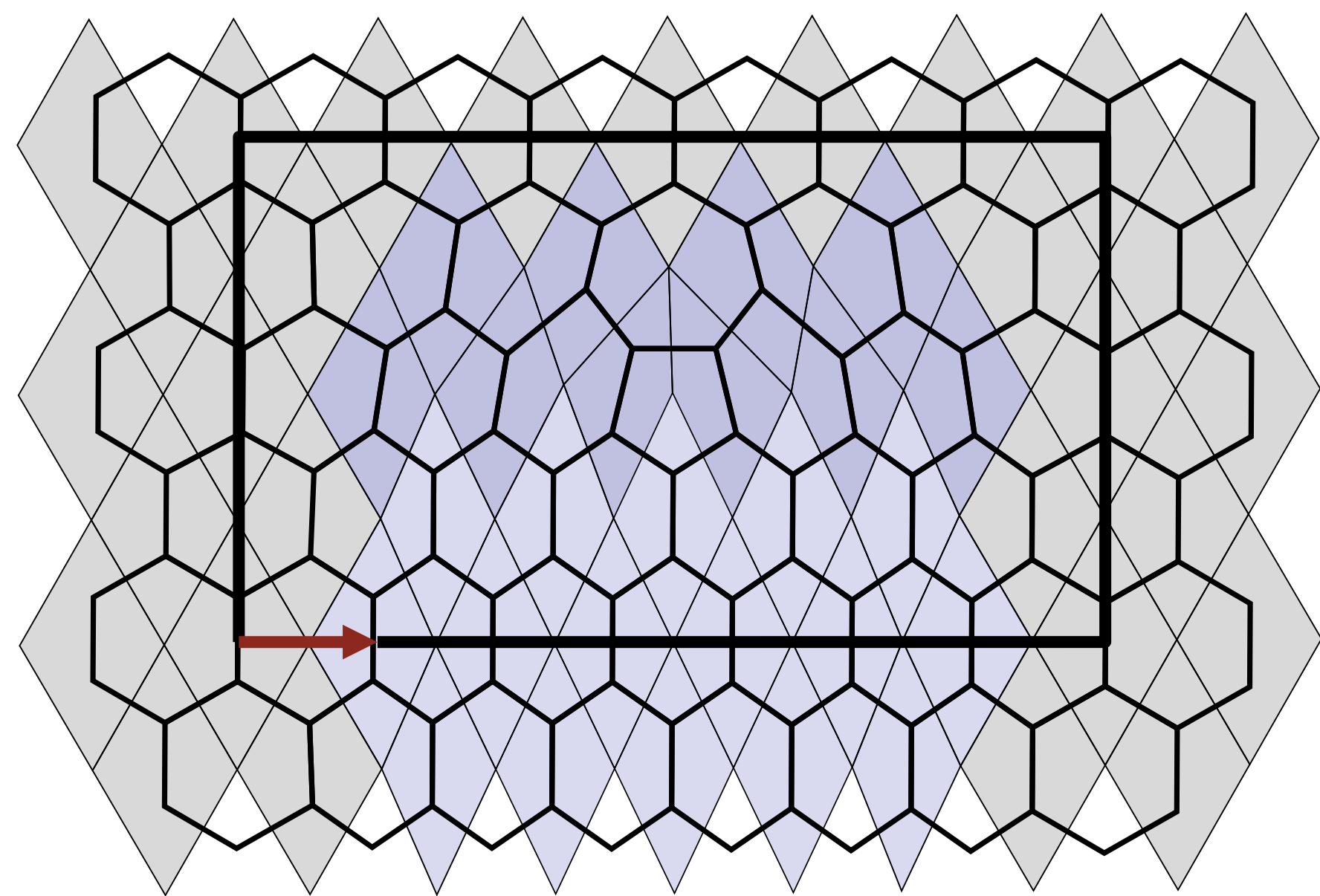
Continuum limit

$$i\partial_i \vec{\phi}(\vec{r}) = i c \vec{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \quad \rho = \vec{\phi}^{\dagger} \vec{\phi}, \quad \vec{j} = -c \vec{\phi}^{\dagger} \vec{\sigma} \vec{\phi}$$

$$\Downarrow$$

$$i\gamma^{\mu} \partial_{\mu} \phi = 0, \quad \gamma^{\mu} = \sigma^{\mu} \sigma^3, \quad \mu \in \{0, 1, 2\}$$

Burgers vector in 5-7 defects



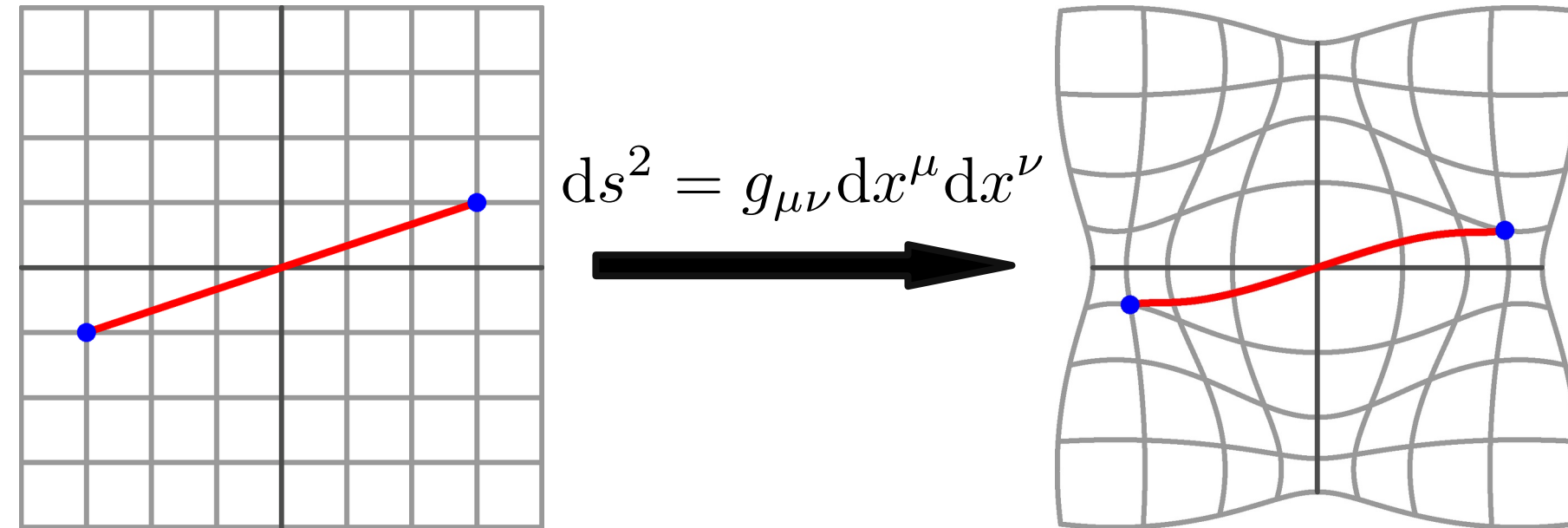
One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = \sqrt{3} \vec{u}_x$

Riemannian Geometry

Riemannian Manifold $\mathcal{M}$ equipped with metric $g_{\mu\nu}$ and connection $\Gamma^{\lambda}_{\mu\nu}$.

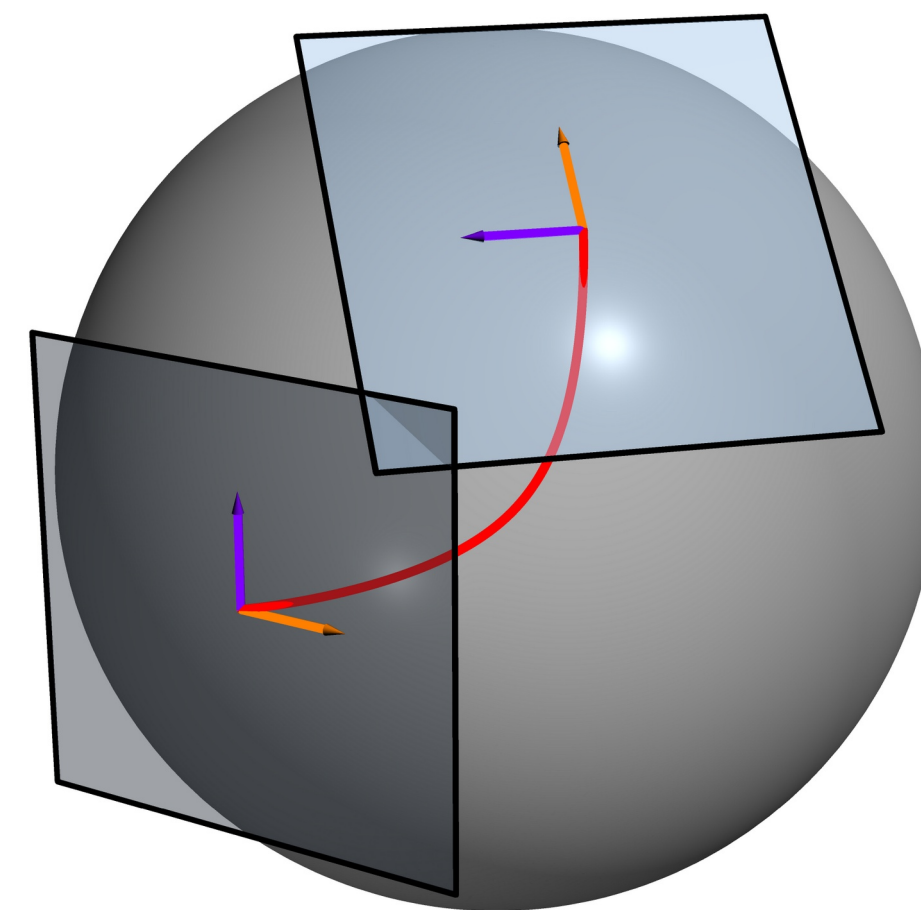
Metric

Defines distances and angles.



Connection

Connects nearby **tangent spaces**, which allows the definition of a derivative on **vector fields** which take on values in the tangent space.



The unique torsion-free, metric compatible connection is called the **Levi-Civita connection**:

$$\{\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} (\partial_{\nu} g_{\rho\mu} + \partial_{\mu} g_{\rho\nu} - \partial_{\rho} g_{\mu\nu})$$

Covariant derivative

Takes into account the structure of the manifold and transforms like a tensor.

$$\nabla_{\mu} v^{\nu} = \partial_{\mu} v^{\nu} + \Gamma^{\nu}_{\mu\lambda} v^{\lambda}$$

Autoparallels

A curve whose tangent vector is parallel-transported along itself.

$$u^{\mu} \nabla_{\mu} u^{\alpha} = 0$$

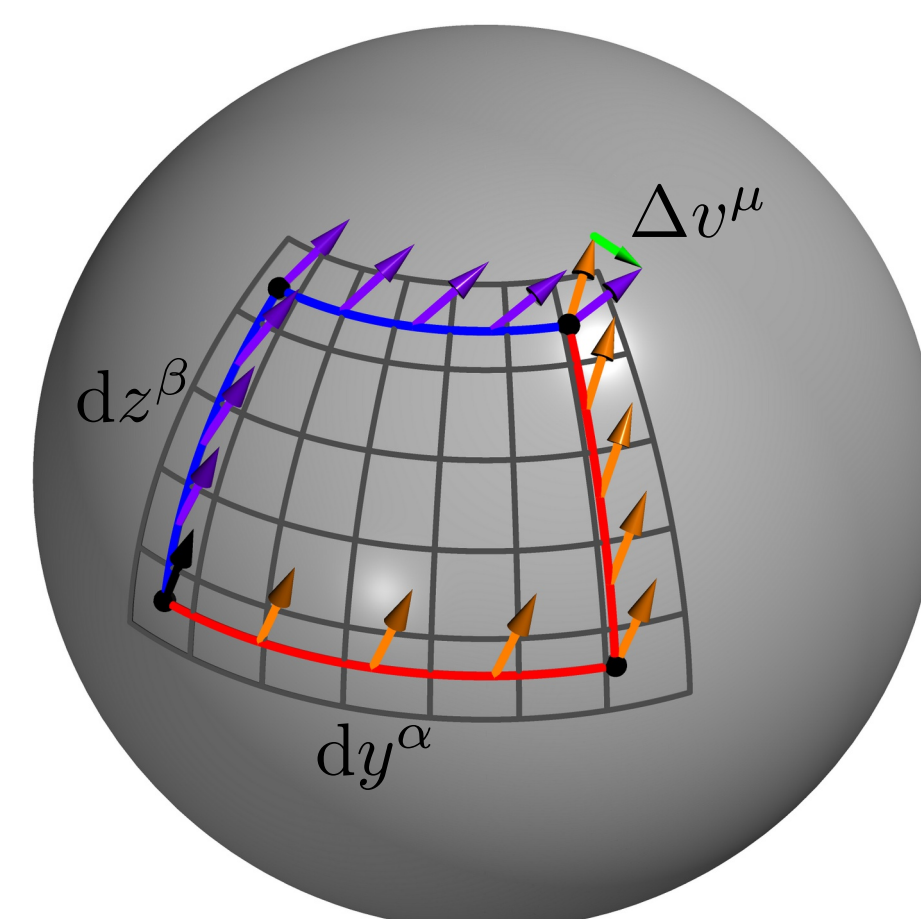
Curvature

The Riemann curvature tensor describes how parallel transport of vectors depends on the path.

$$\Delta v^{\mu} = R_{\alpha\beta}{}^{\mu}{}_{\nu} v^{\nu} dy^{\alpha} dz^{\beta}$$

It can be expressed in terms of the connection:

$$R_{\alpha\beta}{}^{\mu}{}_{\nu} = \partial_{\alpha} \Gamma^{\mu}_{\beta\nu} - \partial_{\beta} \Gamma^{\mu}_{\alpha\nu} + \Gamma^{\mu}_{\alpha\lambda} \Gamma^{\lambda}_{\beta\nu} - \Gamma^{\mu}_{\beta\lambda} \Gamma^{\lambda}_{\alpha\nu}$$



Torsion

In standard GR the torsion is assumed to vanish. It is defined as the antisymmetric part of the connection:

$$T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu}$$

Any general affine connection can be decomposed into a symmetric part, the contorsion and the deformation:

$$\Gamma^{\alpha}_{\mu\nu} = \{\alpha_{\mu\nu}\} + K^{\alpha}_{\mu\nu} + L^{\alpha}_{\mu\nu}$$

Here we assume **metric-compatibility**:

$$\nabla_{\lambda} g_{\mu\nu} = 0 \Leftrightarrow L^{\alpha}_{\mu\nu} = 0$$

The **contorsion tensor** is build from the torsion:

$$K^{\alpha}_{\mu\nu} = \frac{1}{2} (T^{\alpha}_{\mu\nu} - T_{\mu\nu}{}^{\alpha} + T_{\nu}{}^{\alpha}{}_{\mu})$$

TODO:

In depth visual explanation of torsion and how it is analogous to the burgers vector

TODO:

Comparison of models (tight binding vs. autoparallels)
-derivation of scattering angle in torsion picture
-Plot Scattering angle over defects

TODO:

Numerical implementation of the tight binding hamiltonian

- vector matrix system $\rightarrow$ ODE
- unitary timeevolution
- example plot of density before and after scattering

Metric $\leftrightarrow$ torsion mapping in 2D

In 2D we propose a mapping between a **classical** Manifold $\mathcal{M}_g(g_{\mu\nu}, \{\lambda_{\mu\nu}\})$ and **purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k_{ij})$.

In 2D we can always choose coordinates in which the metric is $g_{\mu\nu} = \delta_{\mu\nu} e^{2\phi(x,y)}$.

Also the torsion in 2D can be written using a single covector $T^k_{ij} = b^k \varepsilon_{ij}$

We demand equal **curvature** of both manifolds:

$$R[\mathcal{M}_g] = R[\mathcal{M}_T],$$

This leads to the following solution for the torsion vector b^i as a function of the metric ϕ and an arbitrary scalar field χ :

$$b^i = \varepsilon_{ik} \partial_k \phi + \partial_i \chi$$

For $\chi = 0$ the **autoparallels** of both manifolds are equivalent up to a reparameterization. It can be shown, that the χ field corresponds to local roatations of the frame field.

Conclusion

Except for distances and the problem of different tangent spaces a physical mapping between torsion and metric is possible.

TODO:

Outlook

- Dirac equation on curved background with torsion
- Numeric simulations of this for some torsion fields

TODO:

References